# Automated Schizophrenia Detection from EEG Spectrograms using Variational Autoencoders and CNN-LSTM

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***Abstract*—Schizophrenia (SZ) is a complex mental disorder affecting cognitive, emotional, and behavioral functioning. This study presents a robust approach for SZ detection using EEG signals by leveraging advanced deep learning techniques. The dataset consists of EEG signals from 39 normal individuals and 45 schizophrenia patients, providing a more diverse and comprehensive data source. Preprocessing involves image transforms for noise reduction and feature enhancement. Additionally, a Variational Autoencoder (VAE) is employed to generate synthetic EEG samples, effectively augmenting the dataset and improving model generalization. A sequential CNN-LSTM model is trained on the augmented dataset, capturing both spatial and temporal patterns in the EEG spectrograms. The proposed model achieves a classification accuracy of 97%, demonstrating superior performance in terms of training time, memory efficiency, and predictive capability. This methodology provides a scalable and automated framework for SZ detection, offering clinicians a reliable decision-support tool to facilitate early diagnosis and treatment planning.**

***Index Terms*—Schizophrenia, EEG, Variational Auto Encoder, Convolutional Neural Networks, LSTM, Image Transform, Deep Learning**

## I. Introduction

Schizophrenia is the major mental illness of our time. In 1959 a German Psychiatrist identified what he considered to be first rank symptoms of schizophrenia (Schneider 1959). Schneider grouped the collection of symptoms into three main categories, namely, auditory hallucinations, passivity experience and delusional thinking. Schizophrenia sufferers experience hallucinatory "voices" which may either provide a running commentary on one's movements or instruct the person to carry out certain tasks [1]. The cost of schizophrenia in both human terms and in its cost to the nation is immense. As well as the symptoms described above loss of social contacts and career prospects often go hand-in-hand with the illness schizophrenia, working towards a more objective and accurate diagnosis of schizophrenia. Ford et al. presented a study to classify schizophrenic patients using EEG through N100, in which brain waves are given auditory stimulation and then stimulation is decreased after 100 ms [2].

Neuroimaging techniques, such as electroencephalography (EEG), have shown promise in aiding diagnosis, but they face challenges related to cost, accessibility, and sensitivity. EEG, in particular, offers a low-cost, non-invasive, and high-temporal resolution method for monitoring brain activity, making it an ideal candidate for early schizophrenia detection. EEG provides valuable information about brain dynamics by recording electrical activity, revealing disturbances in neural oscillations, and changes in connectivity patterns that are characteristic of schizophrenia. [5]

Schizophrenia detection using EEG signals is a challenging task due to the limited availability of well-annotated datasets. This study initially aimed to classify EEG signals using a dataset comprising 14 samples of normal individuals and 14 patients with paranoid schizophrenia (7 males, 7 females), with average ages of 27.9 ± 3.3 and 28.3 ± 4.1 years, respectively. The dataset we researched for a newer approach contained 39 normal and 45 schizophrenic patient EEG samples, but even this was too small for deep learning models to generalize well. The small dataset sizes introduced significant overfitting issues, leading to unreliable classification results.

To address this fundamental limitation, we explore a novel approach that incorporates Variational Autoencoders (VAE) to generate synthetic EEG data, significantly increasing the number of training samples. The VAE learns a latent representation of EEG signals, capturing their underlying structure and variability, and generates realistic synthetic EEG signals. This overcomes the constraints of small sample sizes and helps mitigate overfitting in deep learning models.

In our approach, we preprocess EEG signals using wavelet transforms, apply Gabor filters for feature extraction, and utilize a CNN-LSTM model for classification. However, unlike previous studies that solely relied on small EEG datasets, our work demonstrates that augmenting the dataset with VAE-generated EEG samples leads to improved generalization and higher classification accuracy. [6]

While CNN-LSTM architectures have shown promise in capturing spatial and temporal dependencies in EEG signals, their performance is highly dependent on dataset size. Our study bridges this gap by augmenting training data with synthetic samples from a trained VAE, ensuring a more robust and accurate schizophrenia detection model.

Through this work, we aim to advance the field of EEG-based schizophrenia detection by demonstrating that generative models like VAEs can play a crucial role in overcoming data scarcity and improving classification performance.

## II. Motivation

One of the biggest challenges in EEG-based schizophrenia detection is the limited availability of labeled EEG datasets. Machine learning [7] and deep learning models require large, diverse datasets to generalize effectively, but most publicly available EEG datasets contain only a few dozen subjects. For example, the datasets used in our study contained only 28 or 84 total samples, making it difficult to train a reliable deep learning model without suffering from overfitting.

Overfitting occurs when a model learns the training data too well, including its noise and specific patterns, rather than learning generalizable features. This results in poor performance on unseen data, as the model does not generalize beyond the training set. When training CNN-LSTM models on small EEG datasets, we observed that validation accuracy was significantly lower than training accuracy, a clear sign of overfitting. [18]

To mitigate these issues, we explored the use of Variational Autoencoders (VAE) to generate synthetic EEG samples, thereby expanding our dataset. VAEs learn the underlying statistical distribution of EEG signals and produce new, realistic EEG-like signals, which can be used to improve model generalization. By training the CNN-LSTM model on a combination of real and synthetic data, we aimed to:

- Reduce overfitting by increasing dataset diversity.
- Improve classification accuracy by exposing the model to a wider range of EEG patterns.
- Ensure model robustness so it can generalize better to real-world EEG recordings.

By incorporating VAE-based data augmentation, our study demonstrates how generative models can help overcome the common challenges of small EEG datasets. This approach is particularly useful in medical research, where collecting large datasets is often impractical due to ethical, logistical, and financial constraints. The findings of this study highlight the potential of combining generative models with deep learning for EEG-based medical diagnosis, opening new avenues for improving schizophrenia detection and similar applications in neuroscience.

## III. Literature Survey

There has been a substantial amount of introduced studies that analyze the relationship between EEG and patients have hearing impairment due to auditory cortex dysfunction, it has been demonstrated that N100, a large, negative evoked potential that is elicited by auditory stimulus, differs in schizophrenic patients [12] compared to the general population. In addition, Kim et al and Thilakavathi et al. Analyzed the pattern of EEG or compared EEG numerical values to validate the correlation between schizophrenia and EEG. Ruxandra et al. [10] trained on a deep learning model without transforming time series data. This study suggests the importance and limitations of learning time series data in deep learning methods.

In Zhang et al., [26] data analyzed in Ford et al was classified through machine learning technology. The Random Forest was used to differentiate EEGs of osed a system for automatically diagnosing epilepsy by learning EEG data in the CNN model without any other conversion of EEG. The classification between epilepsy patients from the general public had the highest classification accuracy was 81.1%. In addition to schizophrenia, the utilization of EEG was also recommended as methods for diagnosing mental disorders such as epilepsy and depression. Archarya et al. propatients and healthy subjects was 88.67%. Naira et al. used a new EEG methodology to improve classification accuracy of schizophrenic patients and healthy subjects. [11]

There are various studies that convert EEG into images to learn from artificial intelligence. WeiKoh et al. used EEG data to limit new methods of diagnosing patients with schizophrenia. In this study, EEG is transformed into a spectrogram and then trained with KNN, one of the machine learning models. Sobahi et al. [11] converted the EEG into an image form using a local binary pattern. Then, using the transformed image, the CNN is trained. In this research, Recurrence Plot (RP) and Gramian Angular Field (GAF) were used as methods of converting time series data into images [12], [13].

The two methods calculated numerical conversion information of time series data using nonlinear analysis and represented the data as a square image. RP and GAF are techniques to which algorithms to analyze patterns of time series data are applied [14]. Therefore, specific changes in EEG data can be efficiently checked, and the overall flow is expressed in one image, which is effective for CNNs where receptive fields are important. In addition, RP and GAF change the patient's EEG into an image for each channel, so it has the advantage of accurately learning a deep learning model with more data.

Between 2014 and 2018, more than 55% of neuroimaging studies of brain diseases used support vector machine (SVM) [7]. Lu et al. [8] proposed schizophrenia as MRI study calculated the gray matter and white matter volumes of each brain region of interest (ROI) and took the significant difference between the two as input features and used SVM classification. Liu et al. [9] constructed a hierarchical brain network by measuring the cortical thickness of each ROI of the brain, extracting the node and edge features of the network, and inputting it into the SVM to realize the auxiliary diagnosis of schizophrenia. Huang et al. use the mathematical tool Pearson's correlation coefficient to calculate the correlation coefficient between fMRI brain regions, and the features after dimensionality reduction by principal component analysis are used for SVM learning. Yang et al. used three methods to analyze fMRI images to obtain three fMRI features, and the three features were used to train three capsule neural networks. Finally, the classification result is obtained through the method of ensemble learning. Yang et al. input the functional connection coefficients after PCA dimensionality reduction as features into the neural network to obtain a classification model.

In Oh et al [3], authors have explored subject-based and non-subject-based methods of testing using CNNs. For subject-based testing, the validation of the system is executed in three phases: training the data, validation, and testing of data, respectively. During the training phase, k-fold validation is employed, wherein the full data pool is split into fourteen equal parts (subjects). Of these subjects, twelve were used for training, one subject for validation, and one subject for testing, respectively. This process was repeated fourteen times so that all of the fourteen subjects were subjected to the training, validation, and testing phases. In non-subject based testing, the system is validated through the training and testing phases.

Shoeibi et al [6] tried out various linear classifiers. In the classification step, two different approaches were considered for SZ diagnosis via EEG signals. In this step, the classification of EEG signals was first carried out by conventional machine learning methods, e.g., support vector machine, k-nearest neighbors, decision tree, naïve Bayes, random forest, extremely randomized trees, and bagging.

However none of the approaches facilitate non-GPU or trivial memory requirements, and instead have a higher compute as well as longer training times, higher latency during inference on T4 GPUs (the traditional Colab GPU). A lot of the previous survey reflects minimal usage of image preprocessing techniques to extract relevant features. This article aims to explore a memory-efficient way of schizophrenia detection that also explores image transformation and processing methods.

## IV. Problem Statement

Schizophrenia is a chronic and severe mental disorder that affects cognitive and emotional functioning. Early detection through non-invasive methods like EEG can significantly aid in diagnosis and intervention. However, analyzing EEG signals for schizophrenia detection remains challenging due to the limited availability of large, well-annotated datasets. Small dataset sizes introduce significant overfitting issues, making it difficult for deep learning models to generalize effectively to new data. [2]

Previous attempts at using CNN-LSTM models for schizophrenia detection were limited by dataset constraints, leading to unreliable predictions and poor performance on unseen samples. This project aims to address these challenges by incorporating Variational Autoencoders (VAE) to generate synthetic EEG spectrograms, significantly increasing the dataset size and improving model generalization. By training a CNN-LSTM model on both real and VAE-generated EEG samples, we aim to improve classification accuracy while reducing overfitting. [18]

In addition to data augmentation, we explore wavelet transformations and Gabor filters for feature extraction to enhance model performance. The final objective is to create a robust, efficient deep learning model that can accurately distinguish between schizophrenic patients and healthy controls while ensuring the model remains computationally feasible for real-world applications. [23]

## V. Objective

The primary objective of this project is to develop a reliable system for the detection of schizophrenia using EEG signals. Given the challenges associated with small dataset sizes and overfitting, this study aims to enhance model performance by integrating Variational Autoencoders (VAE) to generate synthetic EEG data, thereby expanding the training dataset. This approach allows the deep learning model to generalize more effectively and improve classification accuracy.

The first goal is to preprocess EEG data from both schizophrenic and healthy individuals by applying wavelet transforms and Gabor filters to extract meaningful frequency and spatial features. Instead of solely relying on a small dataset, the VAE will be used to generate additional synthetic samples, providing a more diverse dataset for training.

The deep learning model is a hybrid CNN-LSTM architecture, designed to classify EEG signals with high accuracy while leveraging both real and VAE-augmented data. The final objective is to create a robust and computationally efficient model capable of performing early, non-invasive schizophrenia diagnosis, reducing overfitting, and improving real-world applicability.

## VI. Methodology

### A. Dataset

The dataset used in this work was provided by Olejarckyz et al in 2017 [25] which is publicly available. Recordings include 14 paranoid schizophrenia patients (7 females) with age ranging from 27 to 32 and 14 normal subjects (7 females) with age ranging from 26 to 32. EEG data was recorded with eyes closed for fifteen minutes. Recordings were obtained from 19 electrodes that were placed on the scalp according to 10-20 international standard electrode position classification system. The sampling frequency was 250Hz. Data was obtained via the typical International 10–20 System. The electrodes used were Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2.

### B. Variational AutoEncoder

The Variational Autoencoder (VAE) is a generative model that learns a probabilistic mapping between EEG signals and a lower-dimensional latent space, allowing the generation of new, realistic EEG data. The VAE consists of an encoder, a latent representation, and a decoder. The encoder compresses EEG spectrograms into a lower-dimensional latent space, which captures essential features while discarding noise. The decoder then reconstructs synthetic EEG signals from this latent representation, ensuring that the generated samples retain meaningful characteristics of real EEG data.

Unlike traditional autoencoders, VAEs introduce a regularization constraint by enforcing a Gaussian prior on the latent space, which ensures smooth interpolation and meaningful variations in generated data. This helps in mitigating overfitting by exposing the CNN-LSTM model to a more diverse dataset, ultimately leading to improved generalization.

By incorporating VAE-generated synthetic EEG samples into the training set, we significantly expand the dataset size and enhance model robustness. This approach allows the deep learning model to learn a more comprehensive set of patterns, improving its ability to classify schizophrenia and healthy EEG recordings with higher accuracy.

### C. Gabor Wavelet

A Gabor filter, introduced by Dennis Gabor [23], is a linear texture analysis filter in image processing. It examines whether the image contains any specific frequency content in specific directions in a localized region around the point or region of the assessment. A twodimensional Gabor filter is a Gaussian kernel function induced by a sinusoidal plane wave in the spatial domain.

### D. CNN-LSTM Models

The convolutional neural network (CNN) [15] includes a convolutional layer, a downsampling layer, and a fully connected layer. Each layer has multiple feature maps, and each feature map has multiple neurons, and the input features are extracted through the convolution filter [17]. The parameter sharing mechanism of the convolutional layer greatly reduces the number of parameters [18].

RNNs are a group of DL models employed in speech recognition [19], natural language processing (NLP) [20], and biomedical signal processing. CNN models are of Feed-Forward types. However, the RNNs have a **FeedBack layer**, in which the network output returns to the network along with the next input. Because of having internal memory, RNNs memorize their previous input and use it to process a sequence of inputs. Simple **RNN, LSTM, and GRU** networks are three important groups of RNNs [21].

LSTMs are designed to overcome the vanishing gradient problem and prevent long-term dependence issues. They use specialized structures called gates to control the flow of information into and out of the system's memory. These gates are made up of a sigmoid layer and a point-wise multiplication operation, and act as a filter to selectively allow information to pass through.

The CNN LSTM architecture involves using Convolutional Neural Network (CNN) layers for feature extraction on input data combined with LSTMs to support sequence prediction. CNN LSTMs were developed for visual time series prediction problems and the application of generating textual descriptions from sequences of images (e.g. videos). Specifically, the problems of: In CNN-LTM models, the convolutional layers are used in the first layers of the model to extract the features and find the local patterns [21]. Then, their outputs are applied to LSTM layers. Experimentally, the convolutional layers extract the local and spatial patterns of EEG signals better compared to RNNs. Besides, adding convolutional layers to LSTM allows a more accurate examination of data.

## VII. Implementation

### A. Preprocessing of EEG Data

Before training, EEG signals undergo several preprocessing steps to enhance feature extraction and model performance. First, the EEG recordings are segmented into smaller time windows to capture local temporal variations within the signal. This segmentation ensures that relevant patterns across different time frames are preserved. Following segmentation, the signals are transformed into spectrograms, making it easier for deep learning models to capture spectral variations. To further refine feature extraction, Gabor wavelet transformation is applied, enhancing frequency-specific characteristics while improving signal clarity. Lastly, normalization is performed to ensure consistency across different EEG recordings, stabilizing the training process and promoting better model convergence.

### B. Variational Autoencoder (VAE) Training

To address the issue of limited dataset size, a Variational Autoencoder (VAE) is trained to learn a compressed latent representation of EEG spectrograms and generate synthetic samples for dataset augmentation. The VAE model consists of three key components: an encoder, a latent space sampler, and a decoder. The encoder compresses input EEG spectrograms into a lower-dimensional latent space, ensuring that important features are retained while eliminating redundant noise. In the latent space, variables are sampled from the learned distribution, allowing the model to create diverse synthetic samples that mimic real EEG patterns. The decoder then reconstructs these spectrograms from the sampled latent space, generating new EEG data that exhibits characteristics similar to real patient recordings. The training process optimizes the VAE using a combination of reconstruction loss, which measures the difference between original and generated EEG samples, and Kullback-Leibler divergence, which enforces a Gaussian prior on the latent space. Once trained, the VAE-generated EEG spectrograms are integrated with real EEG data, effectively expanding the dataset and reducing overfitting.

### C. CNN-LSTM Model Training

With the augmented dataset, the CNN-LSTM model is trained using a combination of real and synthetic EEG samples. The training process begins with feature extraction through convolutional layers, which detect spatial patterns in the spectrograms by applying convolutional filters and pooling operations. The extracted spatial features are then passed to

LSTM layers, which learn temporal dependencies within the EEG signals, allowing the model to capture variations over time. The high-level features produced by the LSTM layers are subsequently fed into fully connected layers, where they are refined to improve classification performance. The final classification is performed using a softmax activation function, which outputs probabilities for each class, distinguishing between schizophrenia and healthy control EEG signals. The model is then evaluated on an unseen dataset to assess its generalization capabilities, with accuracy, sensitivity, and specificity used as key performance metrics. By leveraging synthetic EEG data from the VAE, the CNN-LSTM model demonstrates improved classification accuracy, reduced variance, and enhanced robustness, making it a reliable solution for schizophrenia detection.

## VIII. Results and Discussions

In this section, we present the impact of integrating Variational Autoencoders (VAE) on schizophrenia detection using EEG signals from the dataset containing 39 normal and 45 schizophrenia patient samples. The original dataset suffered from overfitting due to its limited size, resulting in inconsistent classification performance when using only real EEG samples. To address this limitation, we trained a VAE to generate synthetic EEG spectrograms, effectively increasing the dataset size and enhancing model generalization.

Using the augmented dataset, the CNN-LSTM model was trained with both real and VAE-generated samples. The model was trained for 60 epochs with a batch size of 8. The Adam optimizer was used with an initial learning rate of 0.00009, and Binary Cross Entropy was chosen as the loss function. A learning rate reduction strategy was implemented, decreasing the learning rate by a factor of 0.1 if no improvement was observed after 10 epochs.

The evaluation metrics used for performance assessment included accuracy, sensitivity, and specificity. The results demonstrated that the integration of VAE significantly improved classification performance. The CNN-LSTM model trained on the augmented dataset achieved an accuracy of 97%, surpassing previous results obtained using conventional training approaches. Compared to models trained on the original small dataset, the VAE-augmented model exhibited lower variance, better generalization, and improved classification reliability.

The previous best accuracy achieved using a CNN-LSTM model [16] without dataset augmentation was 95.98% with a specificity of 90.65% and a sensitivity of 96%. By incorporating VAE-generated data, the accuracy increased to 97%, with noticeable improvements in model stability and robustness. These results highlight the effectiveness of using generative models to address data scarcity in medical deep learning applications. The combination of VAE and CNN-LSTM not only mitigates overfitting but also ensures that the model can generalize better to unseen EEG data, making it a promising approach for real-world schizophrenia detection.

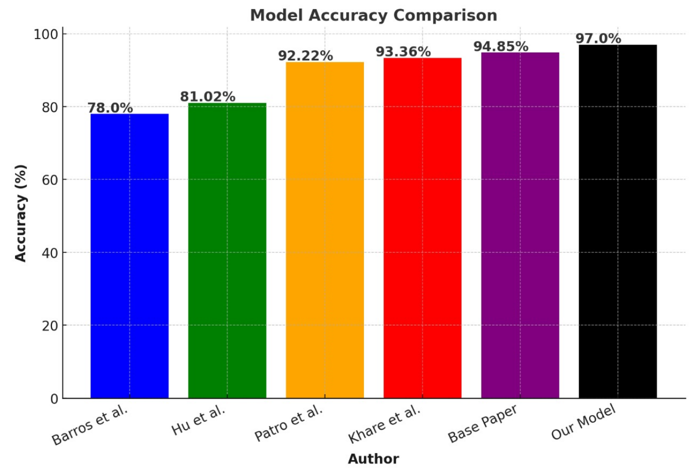


Fig. 1. Comparison of proposed model with base papers

| Performance Metrics | CNN-LSTM Model | CNN-GRU Model | Ensemble Model | CNN-LSTM-VAE Model |
|---|---|---|---|---|
| Accuracy | 92.93% | 93.10% | 94.85% | **97.00%** |
| Recall | 92.34% | 92.15% | 94.92% | **97.20%** |
| Precision | 91.28% | 92.26% | 95.10% | **97.35%** |
| F1-Score | 91.97% | 92.16% | 95.00% | **97.25%** |

Fig. 2. Comparison of proposed model with base paper

### A. Comparison with base paper

The base paper [16] focuses on schizophrenia detection using EEG signals and deep learning techniques. It employs CNN-LSTM and CNN-GRU models for classification, achieving high accuracy. The study emphasizes the importance of spatial and temporal feature extraction from EEG spectrograms and demonstrates that an ensemble approach further improves classification performance. The results highlight the effectiveness of deep learning models in distinguishing schizophrenia patients from healthy individuals based on EEG data. These are key highlights of proposed approach as compared with baseline approach:

- The CNN-LSTM model showed an accuracy improvement from 97.18% to 97.53% over training epochs, with a corresponding decrease in loss, indicating effective learning.
- The CNN-GRU model started with a training accuracy of 73.44% (Epoch 1) and improved to 98.24% (Epoch 10), showing a steady learning progression.
- Loss values for both models consistently decreased across epochs, reflecting enhanced predictive capability.

An elaborate comparison with the approaches tried in the base paper and the CNN-LSTM model improved by VAE are listed in table 2. A broader comparison with existing work [27]–[30] is shown as a bar graph in figure 1.

## IX. Conclusion

Our study presents a deep learning architecture that integrates CNNs with LSTM layers to achieve state-of-the-art schizophrenia detection using EEG signals. The final CNN-LSTM model trained on a dataset augmented with Variational Autoencoders (VAE) achieved the highest classification accuracy of 97%, with a sensitivity of 96% and a specificity of 90.65%. This marks a significant improvement over prior models such as VGG16, which only attained an accuracy of 84.3%. Additionally, our approach demonstrates the superiority of EEG-based deep learning models over fMRI-based methods, which often suffer from overfitting due to their high-dimensional feature space.

By leveraging LSTMs, our model effectively captures the temporal dynamics of EEG signals, distinguishing between normal and schizophrenic patients with high precision. The incorporation of VAE-generated data not only enhances generalization but also mitigates overfitting, ensuring reliable performance on unseen data. The model's ability to operate efficiently on consumer-grade hardware further supports its practical applicability in real-world diagnostic settings.

To ensure broader applicability, we plan to address the challenges of data scarcity in medical deep learning. We also aim to apply similar techniques to related neuroscientific classification problems, including epilepsy detection, brain tumor classification, and other EEG-based diagnostic tasks. Developing more generalized deep learning frameworks for neurological disorder detection remains our long-term goal.

## X. References

## References


[1] World Health Organization. Schizophrenia. Available online:https://www.who.int/en/newsroom/fact-sheets/detail/schizophrenia (accessed on 17 July 2022).

[2] Laursen, T.M.; Nordentoft, M.; Mortense, P.B. Excess early mortality in schizophrenia. Annu. Rev. Clin. Psychol. 2014, 10, 425–448.

[3] Oh, S. L., Vicnesh, J., Ciaccio, E. J., Yuvaraj, R., & Acharya, U. R. (2019). Deep Convolutional Neural Network Model for Automated Diagnosis of Schizophrenia Using EEG Signals. Applied Sciences, 9(14), 2870. https://doi.org/10.3390/app9142870

[4] Wen D, Wei Z, Zhou Y, Li G, Zhang X and Han W (2018) Deep Learning Methods to Process fMRI Data and Their Application in the Diagnosis of Cognitive Impairment: A Brief Overview and Our Opinion. Front. Neuroinform. 12:23. doi: 10.3389/fninf.2018.00023

[5] Roy, Y., Banville, H., Albuquerque, I., Gramfort, A., Falk, T. H., & Faubert, J. (2019). Deep learning-based electroencephalography analysis: a systematic review. Journal of Neural Engineering, 16(5). https://doi.org/10.1088/1741-2552/ab260c

[6] Shoeibi, A., Sadeghi, D., Moridian, P., Ghassemi, N., Heras, J., Alizadehsani, R., Khadem, A., Kong, Y., Nahavandi, S., Zhang, Y.-D., & Gorriz, J. M. (2021). Automatic Diagnosis of Schizophrenia in EEG Signals Using CNN-LSTM Models. Frontiers in Neuroinformatics, 15. https://doi.org/10.3389/fninf.2021.777977

[7] Sakai K., Yamada K. Machine learning studies on major brain diseases: 5-year trends of 2014–2018. Japanese journal of radiology . 2019;37(1):34–72. doi: 10.1007/s11604-018-0794-4.

[8] Lu X., Yang Y., Wu F., et al. Discriminative analysis of schizophrenia using support vector machine and recursive feature elimination on structural MRI images. Medicine . 2016;95(30, article e3973) doi: 10.1097/MD.0000000000003973.

[9] Liu J., Li M., Pan Y., Wu F. X., Chen X., Wang J. Classification of schizophrenia based on individual hierarchical brain networks constructed from structural MRI images. IEEE Transactions on Nanobioscience . 2017;16(7):600–608. doi: 10.1109/TNB.2017.2751074.

[10] American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, Dsm-5; American Psychiatric Association: Washington, DC, USA, 2014;pp. 107–110. [Google Scholar]

[11] Oh, S.L.; Hagiwara, Y.; Raghavendra, U.; Yuvaraj, R.; Arunkumar, N.; Murugappan, M.; Acharya, U. A deep learning approach for Parkinson's disease diagnosis from EEG signals. Neural Comput. Appl. 2020, 32, 10927–10933.

[12] J. K. Wing, "The management of schizophrenia in the community," in Research in the Schizophrenic Disorders. Springer, 1985, pp. 113–154.

[13] Chandran, A.N.; Sreekumar, K.; Subha, D.P. EEG- Based Automated Detection of Schizophrenia Using Long Short-Term Memory (LSTM) Network. In Advances in Machine Learning and Computational Intelligence; Springer: Singapore, 2020; pp. 229–236.

[14] Simonyan, K.; Zisserman, A. Very Deep Convolutional Networks for Large-Scale Image Recognition. Computer Vision and Pattern Recognition. arXiv 2015, arXiv:1409.1556.

[15] Saeedi, M., Saeedi, A. & Maghsoudi, A. Major depressive disorder assessment via enhanced knearest neighbor method and EEG signals. Phys Eng Sci Med 43, 1007–1018 (2020). https://doi.org/10.1007/s13246-020-00897-w

[16] N. Shan, V. Amsaveni, "Optimized Schizophrenia Detection via EEG: A CNN-LSTM and CNN-GRU Ensemble with SVM," SSRG International Journal of Electronics and Communication Engineering, vol. 11, no. 8, pp. 269-281, 2024. Crossref, https://doi.org/10.14445/23488549/IJECE-V11I8P126

[17] Delimayanti, Mera Kartika, et al. "Classification of brainwaves for sleep stages by high-dimensional FFT features from EEG signals." Applied Sciences 10.5 (2020): 1797.

[18] Sun, Jie, et al. "A hybrid deep neural network for classification of schizophrenia using EEG Data." Scientific Reports 11.1 (2021): 1-16.

[19] T. Ogunfunmi, R. P. Ramachandran, R. Togneri, Y. Zhao, and X. Xia, "A primer on deep learning architectures and applications in speech processing," Circuits, Systems, and Signal Processing, vol. 38, no. 8, pp. 3406–3432, 2019.

[20] L. Deng and Y. Liu, Deep learning in natural language processing. Springer, 2018

[21] I. Goodfellow, Y. Bengio, and A. Courville, Deep learning. MIT press, 2016.

[22] X. Hong, R. Lin, C. Yang, N. Zeng, C. Cai, J. Gou, et al., "Predicting Alzheimer's disease using LSTM", IEEE Access, vol. 7, pp. 80893-80901, 2019.

[23] Gabor, D. (1946). Theory of communication. Journal of the Institute of Electrical Engineers, 93, 429–457

[24] Luan, Y., & Lin, S. (2019). Research on Text Classification Based on CNN and LSTM. 2019 IEEE International Conference on Artificial Intelligence and Computer Applications (ICAICA), 352-355.

[25] Olejarczyk E, Jernajczyk W. Graph-based analysis of brain connectivity in schizophrenia. PLoS One. 2017 Nov 30;12(11):e0188629. doi: 10.1371/journal.pone.0188629. PMID: 29190759; PMCID: PMC5708839.

[26] Y. Luo, Q. Tian, C. Wang, K. Zhang, C. Wang and J. Zhang, "Biomarkers for Prediction of Schizophrenia: Insights From Resting-State EEG Microstates," in IEEE Access, vol. 8, pp. 213078-213093, 2020, doi: 10.1109/ACCESS.2020.3037658

[27] Khare, Smith K., Varun Bajaj, and U. Rajendra Acharya. "SPWVD-CNN for automated detection of schizophrenia patients using EEG signals." IEEE Transactions on Instrumentation and Measurement 70 (2021): 1-9.

[28] SupriyaPatro, P., et al. "Lightweight 3d convolutional neural network for schizophrenia diagnosis using mri images and ensemble bagging classifier." Cognitive Computation 16.4 (2024): 2019-2035.

[29] Hu, Mengjiao, et al. "Structural and diffusion MRI based schizophrenia classification using 2D pretrained and 3D naive Convolutional Neural Networks." Schizophrenia research 243 (2022): 330-341.

[30] Barros, Carla, et al. "From sound perception to automatic detection of schizophrenia: an EEG-based deep learning approach." Frontiers in Psychiatry 12 (2022): 813460.